

A Polynomial-Window Obstruction to a Fixed- p Orlicz Weak-Compactness Criterion

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Status

Partial result, likely valid. The result below does not settle the full necessity of the Δ^0 condition in Lefevre–Li–Queffelec–Rodriguez–Piazza [1]. It gives a general obstruction to replacing Δ^0 by weaker growth assumptions when one wants the fixed- p estimate (W).

Source Question

The source paper proves that, if $\Psi \in \Delta^0$ and $X \subset M^\Psi$, then an operator $T : X \rightarrow Y$ is weakly compact if and only if, for some and then for all $1 \leq p < \infty$,

$$\forall \varepsilon > 0 \exists C_\varepsilon > 0, \quad \|Tf\| \leq C_\varepsilon \|f\|_p + \varepsilon \|f\|_\Psi, \quad f \in X. \quad (\text{W})$$

Immediately before Theorem 4, the source asks whether the condition $\Psi \in \Delta^0$ is actually necessary for this characterization.

This growth condition is a strong negation of the Δ_2 condition and it implies that the complementary function $\Phi = \Psi^*$ of Ψ satisfies the Δ_2 condition.

Note that in the following theorem, we cannot content ourselves with $\Psi \notin \Delta_2$ (i.e. $\limsup_{x \rightarrow +\infty} \Psi(\beta x)/\Psi(x) = +\infty$), instead of $\Psi \in \Delta^0$ (see Remark 3 in Section 3). An interesting question is whether the condition $\Psi \in \Delta^0$ is actually necessary for this characterization.

Theorem 4 *Assume that Ψ satisfies the Δ^0 condition, and let X be a subspace of M^Ψ . Then every linear operator T mapping X into some Banach space Y is weakly compact if and only if for some (and then for all) $1 \leq p < \infty$:*

$$(W) \quad \forall \varepsilon > 0, \exists C_\varepsilon > 0, \quad \|T(f)\| \leq C_\varepsilon \|f\|_p + \varepsilon \|f\|_\Psi, \quad \forall f \in X.$$

Remark 1. This theorem extends [12] Theorem II.1. As in the case of $\mathbb{C}^*$ -algebras (see [3], Notes and Remarks, Chap. 15), there are miscellaneous applications of such a characterization.

Remark 2. Contrary to the Δ_2 condition where the constant 2 may be replaced by any constant $\beta > 1$, in this Δ^0 condition, the constant β cannot be replaced by another, as the following example shows.

Example. There exists an Orlicz function Ψ such that:

$$(2) \quad \lim_{x \rightarrow +\infty} \frac{\Psi(5x)}{\Psi(x)} = +\infty,$$

Idea of the Proof

The obstruction is already visible on characteristic functions of small sets. If $\Psi(x)$ behaves like x^q along a subsequence but is much larger than x^p there, then a spike of height x and measure $1/\Psi(x)$ has Orlicz norm one. Its L^p -norm tends to zero, while its L^q -norm remains bounded below. The inclusion into the reflexive space L^q is weakly compact, so these spikes directly contradict the fixed- p version of (W).

Partial Theorem

Theorem. *Let $(\Omega, \mathbb{P})$ be a nonatomic probability space and let Ψ be a strictly increasing Orlicz function. Fix $1 \leq p < q < \infty$. Assume:*

$$x^q \leq K\Psi(x) \quad (x \geq x_0)$$

for some $K, x_0 > 0$, and assume that there are $x_n \rightarrow \infty$ and $C > 0$ such that

$$\Psi(x_n) \leq Cx_n^q, \quad \frac{\Psi(x_n)}{x_n^p} \longrightarrow \infty.$$

Then the inclusion operator

$$J_q : M^\Psi \longrightarrow L^q(\Omega)$$

is weakly compact, but J_q does not satisfy the fixed- p estimate

$$\forall \varepsilon > 0 \exists C_\varepsilon > 0, \quad \|J_q f\|_q \leq C_\varepsilon \|f\|_p + \varepsilon \|f\|_\Psi, \quad f \in M^\Psi. \quad (W_p)$$

Proof. First J_q is bounded. Indeed, if $\|f\|_\Psi \leq 1$, then $\int_\Omega \Psi(|f|) d\mathbb{P} \leq 1$. Splitting into $\{|f| < x_0\}$ and $\{|f| \geq x_0\}$ gives

$$\int_\Omega |f|^q d\mathbb{P} \leq x_0^q + K \int_\Omega \Psi(|f|) d\mathbb{P} \leq x_0^q + K.$$

By homogeneity, $J_q : M^\Psi \rightarrow L^q$ is continuous. Since L^q is reflexive for $1 < q < \infty$, every bounded operator into L^q , in particular J_q , is weakly compact.

We now show that J_q fails (W_p) . Since $(\Omega, \mathbb{P})$ is nonatomic and $\Psi(x_n) \rightarrow \infty$, choose measurable sets A_n with

$$\mathbb{P}(A_n) = \frac{1}{\Psi(x_n)}.$$

Set $f_n = x_n \mathbf{1}_{A_n}$. Then

$$\int_\Omega \Psi(|f_n|) d\mathbb{P} = \Psi(x_n) \mathbb{P}(A_n) = 1,$$

and strict monotonicity of Ψ gives $\|f_n\|_\Psi = 1$. Also

$$\|f_n\|_p = x_n \Psi(x_n)^{-1/p} = \left(\frac{x_n^p}{\Psi(x_n)} \right)^{1/p} \rightarrow 0,$$

whereas

$$\|J_q f_n\|_q = x_n \Psi(x_n)^{-1/q} \geq C^{-1/q}.$$

Choose $0 < \varepsilon < C^{-1/q}/2$. If (W_p) held, then for some C_ε and all n ,

$$C^{-1/q} \leq \|J_q f_n\|_q \leq C_\varepsilon \|f_n\|_p + \varepsilon \|f_n\|_\Psi = C_\varepsilon \|f_n\|_p + \varepsilon.$$

Letting $n \rightarrow \infty$ gives $C^{-1/q} \leq \varepsilon$, a contradiction. $\square$

Corollary. *The factorial Orlicz function used in Remark 3 of the source paper fits the mechanism with $p = 2$ and $q = 3$: along $x_n = n!$ one has $\Psi(x_n) \lesssim x_n^3$ while $\Psi(x_n)/x_n^2 \rightarrow \infty$. Thus its inclusion $M^\Psi \rightarrow L^3$ gives an instance of the obstruction to the fixed- L^2 estimate.*

Verification Notes

The proof is analytic and uses only the Luxemburg norm of indicators and the reflexivity of L^q for $1 < q < \infty$. No computational verification was used.

Limitations and Novelty Search

This packet does not prove that Δ^0 is necessary in the full sense asked by the source. It proves that a fixed lower exponent p cannot control weakly compact inclusions $M^\Psi \rightarrow L^q$ when Ψ has a polynomial window between p and q . Web searches on 2026-07-19 for the source title, “Delta^0 necessary”, “Morse-Transue Delta^0 weakly compact operators”, and Lavergne/Alexopoulos related terms did not locate a later direct answer to the source question.

Human Review Recommendation

Accept as a scoped partial obstruction. Do not count it as a full answer to the source’s Δ^0 -necessity question.

References

- [1] P. Lefevre, D. Li, H. Queffelec, and L. Rodriguez-Piazza, *A criterion of weak compactness for operators on subspaces of Orlicz spaces*, arXiv:0806.4204, submitted 2008-06-25.